Quang-Tri Do

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Summary

Backend Software Engineer with 4+ years of experience designing and delivering scalable, cloud-native systems. Proficient in Python, Java, and Go, with a proven track record of building robust microservices and APIs on AWS and Kubernetes. Passionate about leveraging modern CI/CD practices to deliver high-performance, reliable software in Agile environments.

Experience

Software Engineer @ Phreesia — Ottawa, ON *June 2021 – Present*

- Migrated API calls to Fast Healthcare Interoperability Resources (FHIR) standard after evaluating multiple integration strategies, reducing market costs by over 40% within one year.
- Designed and delivered a real-time cloud communication service in AWS capable of handling 300+ events/second, ensuring system reliability under tight delivery timelines.
- Led end-to-end design and integration of a secure payments workflow, enabling practices to export 3k+ transactions/month seamlessly through third-party APIs.
- Engineered a high-volume patient intake system processing 200k+ daily appointment imports across 12+ markets with improved data accuracy and reliability.
- Prototyped a virtual waiting queue with React, C#, Redis, and WebSocket asynchronous messaging, enhancing patient experience by reducing waiting time in clinics.
- Boosted code reliability by raising unit and integration test coverage to 85%+, reducing production defects and improving CI/CD confidence.

Software Engineer @ Distributel — Ottawa, ON *Feb. 2020 – June 2021*

- Improved flagship application by implementing new C# features and resolving critical bugs, enhancing system stability and service reliability for 5,000+ clients.
- Architected and deployed a unified switch provisioning system consolidating 4 legacy applications, cutting maintenance costs by $100k/year and streamlining operations.
- Spearheaded proof of concept for modern CI/CD practices, reducing deployment time from 1 hour to 15 minutes and accelerating release cycles.
- Owned and maintained three internal applications, implementing fixes and performance improvements that reduced downtime and improved cross-team productivity.

Software Engineer (Co-op) @ BluWave-ai — Ottawa, ON *May 2019 – Nov. 2019*

- Engineered distributed backend and REST services in Python and Go with built-in datastore integration and fault tolerance, improving system scalability and reliability.
- Built a Python-based data ingestion pipeline with 95% uptime, processing 1K+ IoT sensor records daily to support AI-driven energy optimization models.
- Designed and deployed an Azure IoT Edge solution to transfer SCADA system data into internal configuration systems, enabling real-time integration with AI models.
- Achieved 90%+ test coverage across REST API endpoints by implementing unit and integration tests in Go and Python, significantly reducing production defects.

Automation Tools Developer (Co-op) @ Ford Motor Company — Ottawa, ON *May 2018 – Aug. 2018*

- Expanded automated test execution and results analysis, tripling the number of tests executed

Education

University of Ottawa — Ottawa, ON *Bachelor of Applied Science (BASc), Software Engineering*

Projects

Automify – 2nd Place @ Hacking the Future of Work | *HTML, CSS, JavaScript*

- Built an AI-powered tool predicting job automation risk using NLP and labor market datasets. Presented to Trilateral Commission stakeholders and placed 2nd among 20+ teams.

Skills

Languages & Frameworks: Java, C#, Python, Go, .NET, ASP.NET, React, Node.js
Databases: SQL, Postgres, MongoDB, Elasticsearch, Redis
Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, Rancher, Git
Practices: Microservices architecture, REST APIs, CI/CD pipelines, Agile/Scrum, Test Automation, Pub/Sub messaging, Observability (logging, metrics, tracing)